SCHOOL PF INFORMATION AND PHYSICAL SCIENCES

PROCESS ANALYSIS AND PROBLEM SOLVING

INFO6050

Assessment2: Discussion Participation and Contribution

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Campus: Callaghan

Lecture Timings: Monday 3pm-5pm

Lab Timings: Friday 3pm-5pm

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Report Introduction:

I am Hari Krishna Jadda, a full stack software engineer with 5+ years of hands-on experience, in both backend and frontend developer roles. I am originally from India, I am currently pursuing in Master of Information Technology at the University of Newcastle in Callaghan. My professional experience includes working as a Systems Engineer at Tata Consultancy Services where I worked with clients, developed microservices using Java Spring Boot and Kafka, and worked on CI/CD pipelines. As a Senior Systems Engineer at Infosys, I was responsible for RESTful APIs, real-time data systems and collaborating with stakeholders. In addition, while at High Flows IT Solutions as an Associate Software Engineer, I developed backend logic while using Python and Flask, and integrated with frontend elements. Through these experiences, my skills in agile methodologies, secure coding and configuration managment using tools such as Docker and Kubernetes, were refined. Many of these experiences were in high stakes projects for such organizations as MetLife and AIG.

Before starting INFO6050, Process Analysis and Problem Solving, my expectations were centered on gaining practical tools to bridge my technical expertise with broader business analysis skills, as this course aligns with my career shift toward more strategic IT roles in Australia. Currently, my expectations for myself include building a deeper understanding of process documentation, problem solving techniques like brainstorming and mind mapping, and applying these to real world workflows skills that will enhance my ability to identify inefficiencies in software development processes from my past projects. For INFO6050 specifically, I expect it to provide hands on insights into fact finding, stakeholder analysis, and innovative improvements, as outlined in the course content, helping me create high quality business reports and work synergistically in groups. I am genuinely interested in this course, my background in managing client expectations and optimizing systems like performance tuning databases and automating testing draws me to its focus on observing workflows, identifying opportunities, and using techniques like BPR and quality reviews, which I see as directly applicable to advancing my career in cloud application development or API design. Overall, I anticipate this course will expand my problem solving toolkit, linking my technical past to more holistic business outcomes.

Week 1:

Introduction:

This week served as an introduction to INFO6050, covering course objectives, self introductions in the welcome forum, required readings on business analysis, systems thinking, and design thinking, and a forum activity to define these concepts personally. It emphasized building class community and starting to explore how these ideas relate to process improvement in organizations, aligning with the course's focus on tools for observing workflows and identifying opportunities.

Task:

1. In your own words what does a business analyst (BA) mean to you.

A Business Analyst is a key enabler within an organization, serving a critical role in connecting business needs and solutions from a technology perspective. In my definition, a BA is someone who knows the ins and outs of an organization, knows how things can be better, and is able to communicate this need in an actionable way for the development teams. The BA has a long way to go before they become a project manager, or project lead. They are not simply taking notes. Instead, they are taking time to investigate an idea or situation, identify options to move forward, write down next steps, and sometimes even write down things before a project manager gets to develop a plan! The BA's activities include eliciting, analyzing, communicating and validating requirements for changes to business processes, information system and policies. Crucially they will be concerned that the project works are aligned to the strategic goals and deliver benefits for stakeholders. Finally, The Business Analyst is a change agent in the sense of finding opportunities and possibilities for change and improvement of the business environment.

2. In your own words describe what is system thinking using one example.

I see systems thinking as a holistic view of problem solving, that is focused on exploring the relationships and interactions of the different components of a system that come together to create an end result or impact. It looks at the system as a whole, typically not looking at individual parts, it encourages understanding both the interconnectedness and the dynamic nature of the overall system. Systems thinking provides a more realistic view of reality and allows for an understanding of potential long term effects and unintended consequences and is a means of designing sustainable solutions. It is a diagnostic tool and encourages good questions as opposed to jumping to conclusions first. Problems best suited for systems thinking are usually chronic, familiar, and have been difficult to resolve in the past.

Example:

The redesign and relocation project at Johns Hopkins Hospital provides a great example of systems thinking in action. Instead of simply moving departments, the Hospital considered the entire system of the hospital and invited every stakeholder, including doctors, custodians, patients and others, to participate in the design process. Doing so opened up the identification of key interdependencies, which included the critical elements of transportation services and environmental services and their relation to patient experience and overall efficiency of the hospital. The notion of considering the hospital as an entire system resulted in a redesign of the entire process and recruitment, training, and compensation process. this allowed for a more effective and patient centric hospital system, not simply a new building. This provides a compelling example of how a comprehensive understanding of the system, including the neglected pieces in this case, can develop solutions that are not only much more impactful but also sustainable.

3. In your own words describe what is design thinking using one example.

Design thinking, in my opinion, is a human centered, iterative innovation and problem solving process. It is a process that focuses on learning the user needs first, challenging assumptions, and reframing problems in an attempt to seek alternative strategies and solutions that are not obvious with a more traditional problem solving methodology. It is an empathetic mindset, ideation, prototyping, and testing, working to create solutions that are desirable, feasible, and viable. It uses imaginative thinking and a willingness to entertain what might possibly be true, not necessarily what is.

Example:

Consider the example of developing a new mobile banking application. Design thinking would begin with deep empathy for the users with respect to their expenditure habits, what they dislike about their existing bank apps, and aspirations regarding managing money. Interviews, observations, and journey maps could be involved.

With these facts in mind, the design team would set up the underlying issues, for example users find it difficult to keep track of their spending or users are intimidated by complex investment options. They would then come up with a wide range of possible solutions, from simple costing tools to gameified investment capabilities. More promising ideas would go through prototyping, potentially developing low fidelity wireframes or interactive mockups, and then demonstrated to real users. The cycle of receiving feedback and iterating that prototype is how we make sure the final application not only meets user needs, but also provides an intuitive and enjoyable experience, not just functional one.

4. In your view, name one skill you feel is not required for a BA to carry out their profession.

In my view, a skill that is not strictly required for a Business Analyst to carry out their profession effectively is advanced, hands-on programming proficiency. While a foundational understanding of technology and software development lifecycles is crucial for a BA to communicate effectively with technical teams and understand the feasibility of solutions, being able to write complex code or develop applications from scratch is generally outside the core responsibilities of a BA. Their role is to define what needs to be built and why,

rather than how it should be coded. While some BAs might have a technical background and possess programming skills, it is not a universal prerequisite for success in the profession. Their strength lies in analysis, communication, problem-solving, and bridging the gap between business and IT, not in direct software development.

Peer Review I gave to Naga Venkata Saideep Kamuni:

Did the work describe and highlight the chosen topic? Yes

Do you recommend any changes? Yes If

yes, what are they?

You did a good job describing the topics, with clear definitions for a business analyst and systems thinking backed by strong references. However, the section on design thinking doesn't provide a unique definition or its own example, instead only linking it to the previous systems thinking case. To improve, adding a separate, concise explanation of design thinking with a distinct example would make the response more complete and clear.

Peer review I received :

No specific peer review was received for my post during this week

Reflections:

This week, I invested approximately 8 hours overall, 3 hours on the required reading from Chapter 1 of Business Analysis and the integration article, 3 hours researching and drafting the forum post, 1 hour on the peer review, and the remaining time reflecting and connecting ideas. The process felt straightforward but thought provoking. I enjoyed defining systems and design thinking in my own words. For instance, the Johns Hopkins example reminded me of my TCS project with MetLife, where I tuned MySQL databases holistically considering interdependencies with Kafka for real time data much like viewing the entire system to avoid unintended bottlenecks. However, crafting the design thinking example for the mobile banking app was harder, as I drew from my Angular frontend work but had to reframe it user centrically, which challenged my tech focused mindset from roles like building RESTful APIs at Infosys.

Giving the peer review reinforced my learning. Analyzing another's post made me spot gaps in clarity, mirroring how I debug code with JUnit, and it gave me a different perspective on separating systems and design thinking more distinctly. The forum interaction was helpful, fostering a sense of community as per the 8 Ways, like non linear rethinking after feedback. Overall, I wasn't too busy, but it related well to my expectations for INFO6050 enhancing my ability to bridge business and IT, as in my EPOCH Services project where I communicate with stakeholders. It even tied into my life, inspiring me to apply design thinking to a personal hobby project in Python for a budgeting tool. The peer review process wasn't difficult, but it re enforced the value of iterative feedback, similar to Agile sprints in my work history.

Conclusion:

Key outcomes this week included solidifying my understanding of business analysis roles and the interplay between systems and design thinking, while practicing APA referencing and peer collaboration, which built a foundation for deeper process analysis in upcoming weeks.

References:

- Pourdehnad, J., Wexler, E. R., & Wilson, D. V. (2011). Integrating Systems Thinking and Design Thinking. Systems Thinker, 22(9), 2–6.[Retrieved on September 6, 2025] from https://thesystemsthinker.com/wpcontent/uploads/pdfs/220901pk.pdf
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Week 2:

Introduction:

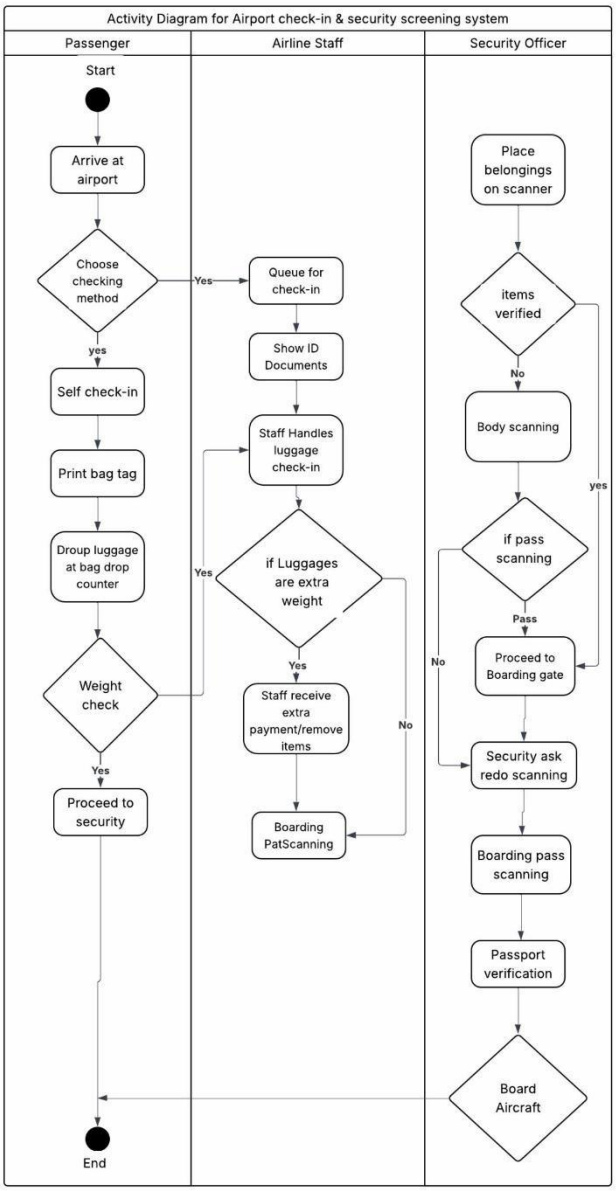
This week centered on the Gidday process to foster class connections, explored further readings on business analysis, and involved a team-based modeling task using Use Case and Activity Diagrams for an airport check-in process case study. The forum activity required documenting our team's work, aligning with the course's focus on process documentation and collaborative problem-solving.

Activity:

Usecase Diagram



Activity Diagram



Peer Review I gave to Vamshi Krishna Daggula:

Hari Krishna Jadda

14 Sep 22:30

Reply from Hari Krishna Jadda

Hi Vamshi,
This is a very strong and clear submission that effectively models the check-in process. One minor comment: in your use case diagram, the action "Update passenger records" is placed before the passenger proceeds to security. and also "Security System" actor is shown with a dotted line. Since it is an external system interacting with the check-in process, it should be represented with a normal, solid line like the other actors.

Peer Review I received from Mohammad Rafid Rabbi :

13 Sep 19:29| Last edited 13 Sep 19:29

Reply from Mohammad Rafid Rabbi

Did the work describe and highlight the chosen topic? Yes

Do you recommend any changes? No

If no, why?
The list captures the key functionalities of an airport check-in and security system highlighting the chosen topic. It successfully identifies the core processes, from the various check-in options and their components. The inclusion of different actors and system functions shows a strong understanding of the scope and the different user interactions within the system. It provides a excellent foundation for building a detailed visual diagram.

Reflections:

This week I spent around 10 hours in total. The readings from Chapters 1-2 of Business Analysis expanded my understanding of process analysis and fact finding techniques, which directly ties into my previous work where I had to elicit requirements from stakeholders to design scalable systems and APIs. For instance, learning about identifying problems in workflows mirrors how I previously analyzed interdependencies in real time data exchanges to ensure alignment with business goals, enhancing my career by equipping me with structured methods to bridge technical solutions with organizational needs more effectively. Techniques on use case and activity modeling provided practical tools for documenting system interactions, which relates to my past experience in mapping out user flows for frontend backend integrations, now I can apply these diagrams to refine process efficiencies in future software engineering roles, making me better at anticipate issues in complex projects.

The forum activity, involving team based creation of Use Case and Activity Diagrams for the airport check-in process, was challenging but rewarding as we collaborated to define actors, boundaries, and flows using draw.io. This hands on exercise built on the readings by applying them to a real-world scenario, similar to how I previously designed workflows for data systems and CI/CD pipelines. It strengthens my career prospects by improving my ability to visualize and communicate system designs in agile environments, ultimately helping me contribute more strategically to cloud application development and API projects. Giving the peer review to Vamshi honed our attention to detail in diagramming conventions, while Mohammad's feedback boosted our team confidence in capturing core processes accurately. Discussing the task with a teammate offered a fresh perspective on subsystem scoping, inspiring a personal project to model a process in Python. Overall, the workload was manageable, and this week's emphasis on modeling and collaboration supports my goal of advancing in roles like full stack or software engineering by formalizing the intuitive process analysis I've used in prior development work.

Conclusion:

Key outcomes this week were mastering UML and Activity diagram creation, strengthening team collaboration, and gaining confidence through peer feedback, setting a solid base for deeper process analysis in upcoming tasks.

References:

- Cadle, J., & Paul, D. (2021). Business analysis techniques: 123 essential tools for success (3rd ed.). BCS Learning & Development Limited.
- Paul, D., & Cadle, J. (2020). Business analysis (4th ed.). BCS, The Chartered Institute for IT.
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Bibliography:

- Navitas Professional. (n.d.). Reflective or learning journal. Retrieved September 12, 2025, from https://sls.navitasprofessional.edu.au/sites/default/files/resource/reflective_or_learning_journal_v1.pdf

Introduction:

This week emphasized undertaking key readings on strategy analysis, commencing the forum activity on PESTLE analysis for a multinational business, and advancing ideas for Assessment 1 while completing any unfinished diagrams from prior weeks. The focus aligned with the course's goals of identifying external and internal factors influencing organizational strategies and adapting to environmental changes.

Activity:

PESTLE analysis for a Mahindra:

Mahindra represents an international business from India which produces tractors along with cars and electric vehicles. Australian rural customers recognize Mahindra as a leading manufacturer of tractors and farm machinery yet, the company works to increase its presence in the car market. The automotive market presents challenges from established brands such as Toyota and Hyundai but Mahindra discovers growth possibilities through customer needs evolution and technological innovation.

Politically, Mahindra operates in Australia based on trade agreements alongside taxation rules and mandatory safety protocols. The government's promotion of electric vehicles through rebates and incentives presents benefits to Mahindra when it launches additional EV models. Australia maintains political stability which provides secure investment conditions but global trade relations between India and other nations continue to affect both market access and operational costs.

Economically, this company faces a limited automotive market which presents intense competition among its competitors. The combination of inflation and rising interest rates together with currency exchange rate fluctuations determines how much buyers can spend on vehicle imports. Rural Australia provides Mahindra with stable revenue through its tractors and farm vehicles which maintain their sales even when economic conditions get worse.

Socially, the purchasing behaviour of Australians now includes sustainability alongside safety and fuel efficiency as main factors for vehicle selection. The current vehicle market demonstrates a distinct transition toward SUVs and electric vehicles particularly within urban areas. The rural customer base of Mahindra exists but the company needs to modify its products to match urban preferences while providing both comfort and environmentally responsible choices to enhance its urban market reputation.

Technologically, the car industry experiences rapid technological advancement through the adoption of electric vehicles alongside smart features and automation. Australia works to

build its electric vehicle charging network which presents a chance for Mahindra to launch its worldwide electric vehicle technology in Australian markets. To challenge major international competitors Mahindra needs to maintain continuous funding for research and development and establish strategic partnerships.

Legally, the automotive company Mahindra needs to follow Australia's demanding automobile regulations including the Australian Design Rules (ADR) which cover safety standards and emission controls. The company must follow consumer protection legislation together with fair trading regulations which affect both vehicle sales and maintenance services. The company's financial standing and public image will suffer if it does not fulfill its required obligations.

Environmentally, The Australian market continues to focus increasingly on both climate change and sustainability issues. Customers now demand businesses to prove their dedication toward reducing emissions while employing environmentally friendly practices. Business operations receive governmental support to adopt renewable energy systems and clean technological solutions. The direction of Mahindra's electric and hybrid vehicles becomes essential for its future development in Australia.

Purpose of a PESTLE analysis and how it could be used by a business?

A PESTLE analysis functions to evaluate external environmental elements which businesses lack control over yet need to react to. Organizations can detect political, economic, social, technological, legal, and environmental elements through this analysis which create both potential advantages and risks. Mahindra utilizes PESTLE analysis to develop Australian market strategies which include EV development because customers show interest and governments back these initiatives.

What did I notice about the business in your own analysis in the previous step?

This analysis revealed that Mahindra achieves its highest success potential in Australia by aligning its operations with sustainable trends and rising electric vehicle consumer preference. The company's tractor products maintain consistent market stability throughout rural zones, yet its automobile division faces intense competition from more dominant brands. The company must earn customer confidence through technological development while forging strategic partnerships with environmental regulations which promote sustainable transportation alternatives.

Peer Reviews I Gave to Premshanker Vanam:

Hari Krishna Jadda

21 Sep 21:48

Reply from Hari Krishna Jadda

Did the work describe and highlight the chosen topic? Yes

Do you recommend any changes? Yes

If yes, what are they?
Your PESTLE analysis for Volkswagen in Australia does a great job of explaining how political, economic, social, technological, legal, and environmental factors affect the company. You should add a precise example by including a Volkswagen ID.4 electric vehicle together with an Australian rule which would make the content more specific. The author should combine specific parts of the societal and technological sections to produce a more focused and straightforward text. The text would become clearer and more focused through these minor changes although the current version delivers the necessary content in a clear manner.

Peer Review I Received from Shashank Sunil Dorepalli:

Shashank Sunil Dorepalli

20 Sep 11:05| Last edited 20 Sep 11:05

Reply from Shashank Sunil Dorepalli

Did the work describe and highlight the chosen topic?
Yes, the PESTLE for Mahindra is clear and covers all the factors well.
Do you recommend any changes?
Yes.
If yes, what are they?
It's a really strong analysis, but some sections could be a bit shorter so the key points stand out more. Also, it would help to link the PESTLE factors more directly to Mahindra's future strategy (like how EV rebates will affect their car plans). Lastly, the references could be tidied up since the McDonald's source is listed twice.
Overall, great job with plenty of detail and good insights!

Reflections:

This week took around 8-10 hours in total. The readings, including Strategy Analysis and the McDonald's PESTLE example, introduced frameworks for scanning external environments, which resonates with my previous work by providing a systematic way to assess market risks tools I can leverage to inform strategic decisions in software projects, like evaluating regulatory impacts on API deployments. The lecture slides on strategic analysis delved into definitions of strategy as long-term direction for organizational advantage and emphasized analyzing external pressures to respond with internal capabilities, which opened my eyes to how unpredictable environments, like AI trends, require proactive scanning skills that will help me anticipate tech shifts in my career, ensuring solutions remain viable amid changes.

Exploring PESTLE in depth during the lecture, including its non mutually exclusive factors and assessing trend impacts, likelihood, and evidence strength, built on the readings by highlighting integration with tools like SWOT for comprehensive strategy execution, this connects to my past experiences where I've had to adapt systems to economic or legal shifts, now empowering me to lead more resilient project planning in full stack roles. The forum activity on creating a PESTLE for Mahindra challenged me to research and synthesize factors

like EV incentives and sustainability demands, drawing parallels to my prior involvement in integrating emerging technologies, this hands on task refined my ability to identify cross category influences, such as how socio cultural shifts toward green tech intersect with technological advancements, positioning me to better evaluate opportunities in competitive tech markets for API development. Giving the peer review to Prem's Volkswagen analysis refined my ability to suggest targeted improvements, much like code reviews I've conducted, while Shashank's feedback on tightening my sections encouraged me to prioritize conciseness in future reports. Interacting with classmates about their chosen businesses sparked ideas on global trade influences, leading me to take down notes for a side exploration of PESTLE in open source tool adoption. This week's emphasis on environmental adaptation aligns with my aspiration to specialize in cloud applications, offering fresh perspectives on legal and technological factors to enhance secure, scalable solutions based on my prior programming background.

Conclusion:

Key outcomes this week were developing a PESTLE analysis, gaining insights from peer feedback, and connecting external analysis tools to strategic planning, building toward stronger problem-solving capabilities in the course. This experience also deepened my appreciation for how environmental scanning can shape innovative tech strategies, motivating me to explore further applications in my learning journey.

References:

- Cadle, J., Paul, D., Hunsley, J., Reed, A., Beckham, D., & Turner, P. (2021). Business analysis techniques: 123 essential tools for success (3rd ed.). BCS Learning & Development Limited.
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Week 4:

Introduction:

This week involved completing readings on business analysis and stakeholder techniques, researching VMOST examples with a focus on Myer and Kmart in Australia, and finalizing Assessment 1 report. The emphasis was on understanding the business context through frameworks like VMOST to guide process analysis models, with the lecture slides reinforcing strategic alignment and practical application in real-world scenarios.

Reflections:

This week took around 8 hours in total. The readings from Chapter 4 of Business Analysis explored the process model, information gathering, emphasizing the importance of business context as a starting point, which ties into my previous work by offering a framework to ensure technical solutions align with organizational goals enabling me to better integrate DevOps practices with broader business strategies in future roles. Optionally reviewing Techniques on stakeholder analysis prepared me for upcoming activities, highlighting interest grids that remind me of prioritizing stakeholder feedback in past API developments, now enhancing my career by improving collaboration in cross functional teams. Researching VMOST through suggested sites like FourWeekMBA and Consuunt revealed its role in vertical alignment, prompting me to analyze how tactics support vision in retail, this sparked interest in applying it to tech firms, like how Amazon's tactics in cloud computing support its customer centric mission. Finding specifics for Myer and Kmart involved digging into investor reports and articles, which was engaging as it showed real contrasts Myer's focus on mergers for scale versus Kmart's innovation in inventory tech relating to my background by illustrating how strategic alignment can optimize processes like

supply chain automation I've worked on. Without a forum this week, I discussed VMOST informally with a classmates, gaining a new view on its use in non profits, which inspired me to outline a personal VMOST for my MIT studies. Overall, the week's self directed research on real world examples aligned with my goal of becoming a cloud application developer, providing tactics to evaluate how objectives like scalability fit into larger visions based on my prior experience in secure coding and microservices.

Conclusion:

Key outcomes this week were completing readings on business process models, researching and noting VMOST for Myer and Kmart, and advancing Assessment 1, fostering a deeper grasp of strategic alignment in analysis. This built my ability to contextualize problems effectively, encouraging ongoing exploration of retail strategies in Australian contexts.

References:

- Cadle, J., Paul, D., & Turner, P. (2021). Business analysis techniques: 123 essential tools for success (3rd ed.). BCS Learning & Development Limited.
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Week 5:

Introduction:

This week included completing readings on stakeholder perspectives and management, engaging in a group-based lab activity to create a Stakeholder Power/Interest Analysis matrix for a case study on a hypothetical offshore wind farm project in the Gulf of Cadiz, Spain, and updating my journal with reflections. The lecture slides provided practical insights into stakeholder identification, analysis techniques like power/interest grids, and their role in business process improvement, emphasizing collaboration in diverse contexts like environmental projects.

Activity:

TASK 1 – STAKEHOLDER ANALYSIS IDENTIFICATION AND OVERVIEW

- Local Fisheries Council.
- Local hotels or tourism developers
- Fishing community & industry
- Greenpeace like environmental NGO's
- Wind farm developers or promoters
- Local residentes of Barbate, Conil, Vejer de la Frontera
- Government or regulators

Classify stakeholders into the powers and interests.

1. High power and high interest - manage closely
2. High power and low interest - keep satisfied
3. Low power and high interest - keep informed
4. Low power and low interest - minimum effort

Stakeholder overview Manage closely:

a. Local Fisheries Council: they can block and support the activities or projects. Also fishing industry directly affected

b. Government: they can approve and cancel the project.

Keep Satisfied:

1. Hotel or tourism developers: they have a local influence, mainly with the local government and local officers. Low tong-term interest compared to fisheries, but its still affected
2. Wind farm developers or promoters: have high power like funding and controlling the projects but interests is economic. Keep informed

1. Fishing community: have low power individually, but highly interested in outcomes.

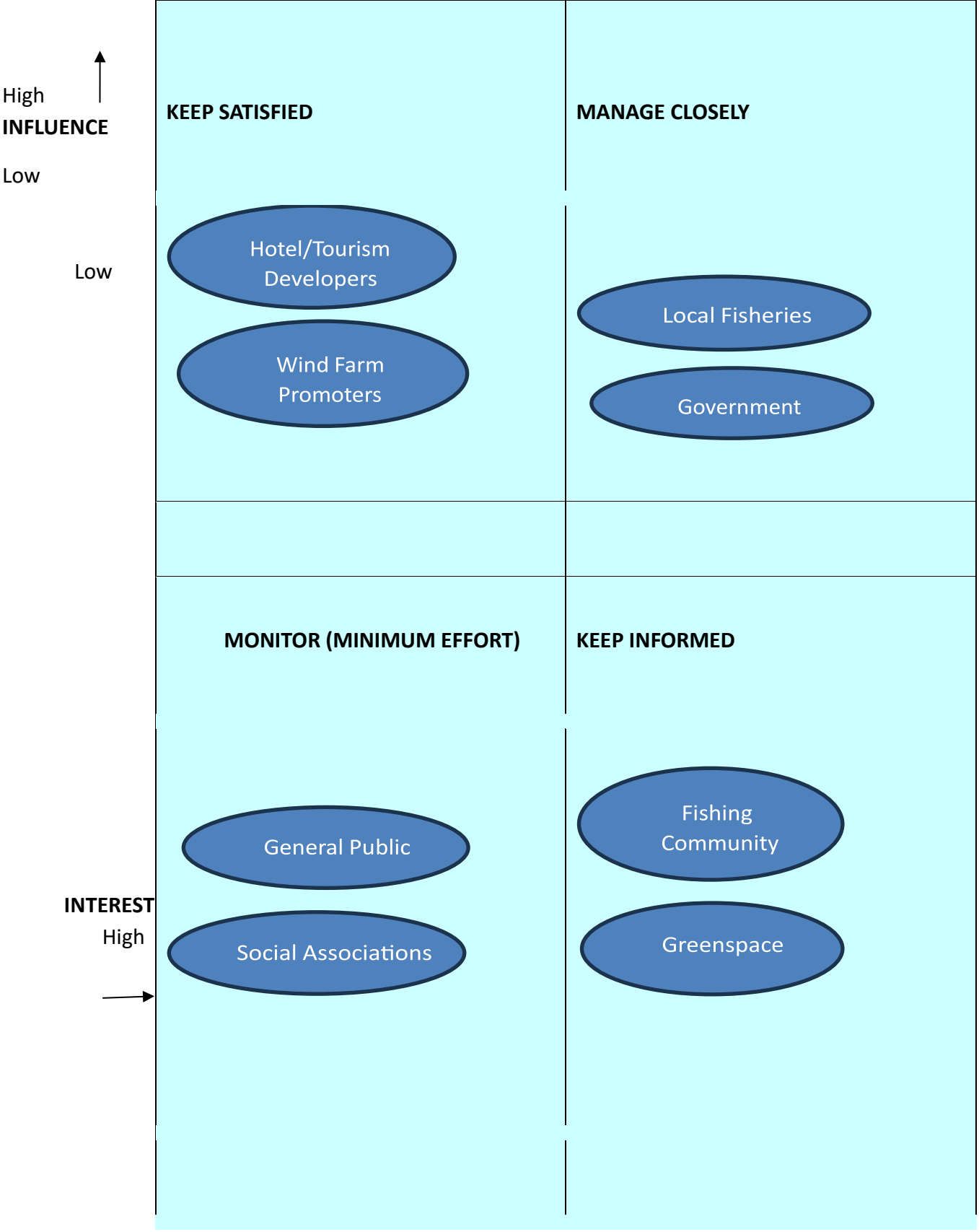
2. Greenpeace: have low direct to stop or cancel the projects, but very high interest in energy and culture impacts.

Minimum effort:

1.General public or residents: low power and also may not be highly impact on project directly.

2. social associations: those who are not directly tied to fishing or tourism have limited influence and interest.

Task 2:



Peer Review we gave to SRI SAI VINUTHNA PINNINTI:

Sankeerth Sappa

3 Oct 15:28| Last edited 3 Oct 15:31

Reply from Sankeerth Sappa

Akshaya Mukkera- c3510621

Sankeerth Sappa - c3508025

Madhavi Latha Tripuram - c3528532

Dhimanshi Jayasena P - c3503338

Hari krishna Jadda - c3511138

1. Was the work describing and pointing out the selected topic?

Yes. The topic is covered; just one thing, you can be more specific in elaborating on trade unions.

2. Do you recommend any changes?

Maybe consider shifting the government body to the Keep Satisfied column, as projects have to inform related government bodies at all times.

Peer Review we got from Shashank Sunil Dorepalli:

Shashank Sunil Dorepalli

3 Oct 15:46| Last edited 3 Oct 15:46

Reply from Shashank Sunil Dorepalli

Did the work describe and highlight the chosen topic? Yes

Do you recommend any changes? Yes

If yes, what are they? Yes, the work discusses and reflects upon the subject matter effectively. I suggest minor alterations to improve clarity in phrasing, support for disciplines for stakeholder categorization, consistency of terminology, enhanced stakeholder matrix, and readibility. The analysis would also be strengthened if you included a brief note about potential stakeholder conflicts or collaborations.

Reflections:

This week took around 9 hours in total. The readings from Chapter 6 of Business Analysis (Paul & Cadle, 2020) explored stakeholder engagement, which aligns with my past work by offering a structured way to prioritize stakeholders enhancing my ability to manage diverse user needs in future full stack projects, like integrating AWS IAM for access control. The lecture slides emphasized stakeholder analysis as a tool for project success, introducing concepts like RACI matrices, which sparked a new realization about clarifying roles, this will

help me streamline communication in multi team environments, drawing from my experience in coordinating requirement gathering.

The group lab activity on the wind farm case study was engaging, as we collaboratively mapped stakeholders using the power grid, revealing nuances like NGOs' influence through advocacy a perspective I hadn't fully considered before but can apply to tech projects with regulatory stakeholders. Shashank's peer review feedback about adding notes on stakeholder conflicts prompted me to rethink potential tensions, inspiring a side note on applying this to a cloud migration project to anticipate user resistance. Our team's peer review to another group, suggesting more detail on trade unions and repositioning government bodies, reinforced the importance of specificity in categorization, mirroring how I've refined access controls in identity platforms. By discussing the grid with my teammates, I was able to identify cultural differences in stakeholder priorities that expanded my view and led me to create a stakeholder map for an app idea of mine. This week's discussion of stakeholder dynamics is relevant for my career focus of becoming a cloud application developer by preparing me for the complex ecosystem that will be found in a real project as I integrate my microservices and DevOps background to help promote collaboration and assist in planning for better solutions.

Conclusion:

Key outcomes this week were completing stakeholder-focused readings, collaborating on a power/interest matrix for the wind farm case study, and exchanging peer feedback, enhancing my understanding of engagement strategies in complex projects. This experience also deepened my ability to anticipate stakeholder interactions, encouraging me to explore conflict resolution in tech project planning.

References:

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Introduction:

This week's focus was on the critical phase of information gathering and the formalization of requirements, covering key investigation techniques and the essential concepts of Functional and Non Functional Requirements, alongside Verification and Validation. The activities included reading relevant chapters on investigation techniques and a team based forum post that applied these concepts to a real world scenario the use of a drone based disaster assessment system to solidify the distinction between verifying the system's build and validating that it meets the user needs. This aligns with the course objective of equipping business analysts with the structured tools necessary to capture precise requirements and ensure solution quality.

Activity:

Group Members:

Madhavi Latha Tripuram - c3528532

Shashank Dorepalli- c3506703

Sankeerth Sappa - c3508025

Dhimanshi Piripun Jayasena - c3503338

Hari Krishna Jadda - c3511138

Akshaya Mukkera- c3510621

Overview of case study:

Business analysts, systems engineers and disaster response specialists are working together to define a drone-based disaster assessment system that will enable quick and secure assessment of regions affected by natural disasters such as floods or earthquakes. Prioritising life-saving rescue and recovery operations, identifying hazards and rapidly comprehending the extent of damage are all made possible by the system's ability to capture and transmit real-time, high-resolution aerial imagery and data.

Core Concepts:

Two fundamental categories of requirements serve as the framework for the creation of the Drone-Based Disaster Assessment System. The system must be able to capture and transmit aerial video, according to the Functional Requirements (FRs). The performance of these functions is specified by Non-Functional Requirements (NFRs), such as the video must be live-streamed in 1080p with less than 2 seconds of latency.

The procedure uses two crucial checks to guarantee quality:

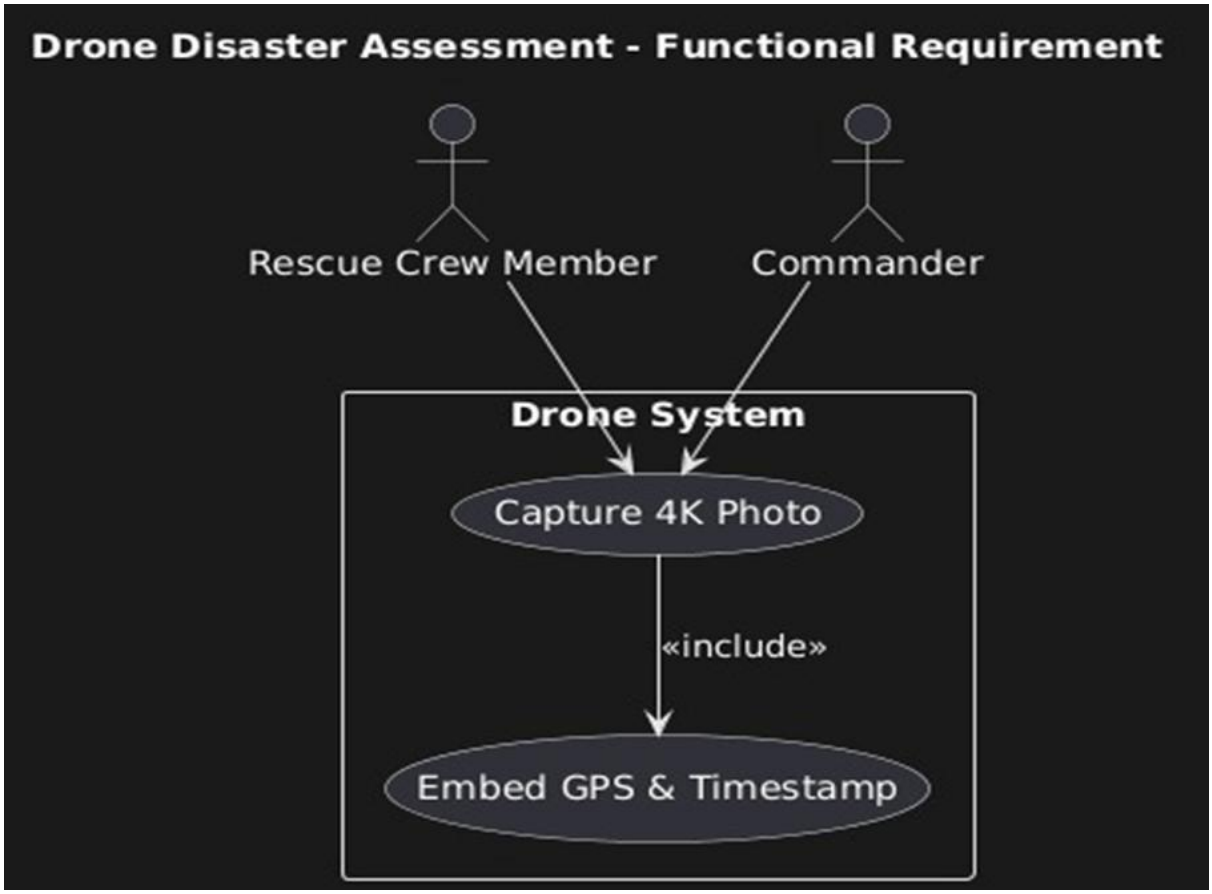
Verification is the technical process of testing the system against its design specifications to ensure that "we are building the product right." By having end users, the Disaster Response Specialists, test the system in real-world situations to make sure it genuinely satisfies their operational requirements, validation, on the other hand, is the pragmatic process of verifying "we are building the right product.

Examples with User Stories and Use Cases:

A powerful way to capture requirements is through User Stories, which describe a feature from the perspective of the person who needs it. Each user story can be linked to one or more Use Cases, which detail the interaction between the user (actor) and the system.

Use Case Diagram:

The following diagram summarizes the key interactions between the primary actor (Rescue Crew Member) and the system.



Functional Requirement: High-Resolution Imaging

High definition photography is the main work feature of the drone. From the user story, "As a Rescue Crew Commander, I would like the drone to capture high-definition 4K photos so that I can visually inspect survivors trapped under debris and structural damage remotely." When the system, upon request from the rescue team member, captures a 4K photo with default metadata, i.e., timestamp and GPS, this requirement is fulfilled.

Verification: Confirming "We Built It Right"

To make sure the system was constructed according to the specifications, verification, a technical quality assurance examination, is carried out by developers and QA engineers. They perform tests in the controlled laboratory setting to verify that the saved image is the precise 3840x2160 pixel resolution, that the software of the flight controller and the hardware of the camera are properly integrated, and that the metadata of the GPS and timestamp are properly embedded into the image file. This process is effectively a box-ticking exercise against the specified technical requirements.

Validation: Confirming "We Built the Right Thing"

Validation is a practical, user-centric check performed by the end-users themselves, such as Rescue Crew Members and Commanders. They evaluate the system in a realistic, simulated disaster scenario, like a training ground with wreckage. Their focus is not on technical specifications, but on operational effectiveness, is the 4K resolution actually sufficient to spot a survivor in a dark crevice or to zoom in and identify critical structural cracks. They also assess practical constraints, such as whether the highresolution file size is manageable for transmission over weak field networks. This direct, hands-on engagement provides crucial feedback on whether the feature truly solves their real-world problem.

Non-Functional Requirements: Battery Life & Live Streaming

A system's critical quality attributes, limitations, and the degree to which it must fulfil its functions are outlined in Non-Functional Requirements (NFRs). In user stories, these are typical acceptance criteria. To maximise search area coverage, for example, a Field Rescue Pilot needs a drone battery that lasts at least 45 minutes, and a Rescue Crew Commander needs a live video feed with a delay of less than 2 seconds so that they can make decisions instantly. The engineering team has specific, quantifiable performance goals since these NFRs are formally documented in use cases for battery monitoring and live-streaming.

Verification: The Technical Performance Check

The process of technically verifying that the system meets its stated NFRs is called verification. The Engineering Team does this in a laboratory. They would take the drone to a standardized wind tunnel with all systems enabled to look at battery life, being certain that

flight time remains at or higher than the 45-minute threshold. They would guarantee end-to-end latency for live streaming using network measurement tools to ensure that it is constantly below 2 seconds. This stage is a pass/fail test against the specified, quantitative requirements.

Validation: The Real-World Effectiveness Check

Validation is the most critical process through which end-users the Commanders rescue team and Pilots test the system under real, operating conditions to determine if it actually meets their needs. This usage often detects major gaps that cannot be foreseen through lab testing. For example, although the battery is tested to endure for 46 minutes within the lab, battling strong winds in the field might drop it to 35 minutes, rendering the search impossible and giving rise to a user-verified demand for a 60minute battery. Similarly, a video latency of 1.5 seconds will pass testing but be unacceptable for precisely flying the drone through a thin window when a user-verified demand for below-500ms latency comes about. This close, hands-on experience ensures that the system is not just technically robust but also resilient and efficient in the wild, dynamic setting of a real disaster.

This case study serves as an example of how inadequate verification on its own is. A system may still have issues in practical settings even after passing all technical tests. It takes direct and continuous interaction with end users (Rescue Crew) through techniques like user story workshops and validation testing in realistic scenarios to create a system that truly meets their needs and can operate reliably in a high-stakes disaster environment.

Peer Review we gave to Botao Ying:

11 Oct 13:17

Reply from Madhavi Latha Tripuram

Group Members:

Madhavi Latha Tripuram - c3528532

Shashank Dorepalli- c3506703

Sankeerth Sappa - c3508025

Dhimanshi Piripun Jayasena - c3503338

Hari Krishna Jadda - c3511138 Akshaya

Mukkera- c3510621

1. **Did the work describe and highlight the chosen topic?** Yes
2. **Do you recommend any changes?** No

3. **If no why?**

Your explanation is very clear and easy to understand. You gave perfect examples for both functional and non-functional requirements.

For the functional part flying to a point, you showed how to check the tech in a lab (verification) and then test it in a real field with users (validation).

For the non-functional part emergency obstacle avoidance, you again clearly showed the difference between lab testing and real-world testing with rescue workers. Your work perfectly shows the difference between "did we build it right?" and "did we build the *right* thing?". No changes are needed. Good job.

Peer Review we got:

No, We have not got any peer reviews.

Reflections:

I dedicated approximately 8 hours this week. 3 hours to the required reading on investigation techniques, 4 hours researching and drafting the team forum post on Verification and Validation, and 1 hour on reflection and connecting the concepts.

The readings on investigation techniques, particularly the suitability of different methods for various situations, were highly valuable. Drawing from my background as a full stack developer at TCS and Infosys, I could immediately relate the concepts of Interviews and Workshops to my prior experience in requirements elicitation. For instance, the distinction between espoused theory and theory in use resonated with past projects where I had to observe users actual workflow to find the true process, rather than relying solely on their documentation or verbal descriptions a classic BA challenge. This week provided a formal framework for what was previously an intuitive part of my job.

The forum activity, which required applying FR, NFR, Verification, and Validation to the drone disaster assessment scenario, was particularly insightful. We defined the drones ability to capture high-definition 4K photos as a clear Functional Requirement. More critically, we focused on Non Functional Requirements such as the drone battery that lasts at least 45 minutes and the live video feed with a delay of less than 2 seconds, which are essential for operational success. Mapping these to the quality checks clarified the difference.

- **Verification** (e.g., checking the code to ensure the image capture function works as designed, or testing the battery life in a controlled lab) is the technical process of ensuring we are building the product right.

- **Validation** (e.g., having Rescue Crew Commanders test if the 4K resolution is actually sufficient to spot a survivor, or seeing if the 45-minute battery life holds up against strong

field winds) is the pragmatic process of confirming we are building the right product for the user's high-stakes environment.

This exercise directly enhances my professional toolkit, particularly for future roles in cloud application development or API design, where defining clear NFRs is paramount to project success. The team collaboration on the post further reinforced the concepts, mirroring the agile sprint review process in my professional past.

Conclusion:

The key outcome of Week 6 was a formalized understanding of the critical requirements engineering concepts: FR, NFR, Verification, and Validation. By applying these to a practical scenario, I have strengthened my ability to not only gather information effectively using structured investigation techniques but also to ensure that the developed solution is both technically sound (verified) and capable of meeting the client's actual business needs (validated), a skill essential for bridging the gap between business and IT.

References:

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Week 7:

Introduction:

This week was centered on the core business analysis topic of Modelling, specifically focusing on the application of Use Case Diagrams and Activity Diagrams to represent system and process flows. The activities included undertaking key readings on these modelling techniques and a hands-on lab exercise where we developed both diagrams for a complex ore crushing process scenario. The main assessment activity for the week was the peer review of a Week 6 team forum post, which reinforced the concepts of Functional and Non Functional Requirements (FR/NFR) and Verification/Validation. The core goal of visual

modeling exists to help stakeholders who possess technical knowledge and those who do not understand complex system interactions.

Reflections:

This week, I invested approximately 7 hours overall, dedicated to reading, diagramming, and the peer review activity.

The theoretical material on Use Case and Activity Diagrams was a valuable re-visitation of UML principles, which I have encountered in my previous roles as a developer. However, the practical application to the ore crushing process scenario highlighted the BA's role in abstracting complexity. For example, creating the Activity Diagram for the entire process, from primary crushing to screening, required a shift in perspective from writing code to visually representing the flow of material and the interaction points of actors like the 'Operator' and 'Quality Control Inspector'. This exercise directly enhances my ability to map out user flows for frontend/backend integrations in a structured, formal way, a skill that is vital for my career advancement in software engineering.

The peer review activity of the Week 6 forum post was a highly effective way to reinforce last week's concepts. By critically analyzing another team's application of FR, NFR, Verification, and Validation to the drone scenario, I gained a deeper appreciation for the precision required in requirements engineering. The review process operated similarly to an agile code review system because it helped me develop a better understanding of modeling standards and the distinction between system requirements and performance standards. The process of iterative feedback collection followed by evaluation and adjustment continues until the desired results become evident. This approach matches my work experience with handling customer expectations and system performance metrics.

Conclusion:

The key outcome of Week 7 was the practical application and reinforcement of UML modelling techniques, specifically Use Case and Activity Diagrams, through the ore crushing scenario. This, combined with the critical peer review of requirements engineering concepts, has significantly improved my ability to formally document, visualize, and validate system processes, strengthening my foundation for bridging technical development with holistic business analysis.

References:

- Paul, D., Yeates, D., & Cadle, J. (Eds.) (2020). Business Analysis (4 ed.). Swindon, United Kingdom: BCS The Chartered Institute for IT.
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Bibliography:

- Nosich, G. M. (2012). Learning to Think Things Through: A Guide to Critical Thinking Across the Curriculum. Pearson Education.
- The Foundation for Critical Thinking. (n.d.). The Elements of Reasoning and the Intellectual Standards. Retrieved from https://www.criticalthinking.org/pages/theelements-of-reasoning-and-the-intellectual-standards/480

Week 8:

Introduction:

This week introduced the crucial concept of Critical Thinking and its application in process analysis and problem-solving. The core activity was a team-based forum post where we applied the SEE-I framework to a scenario, specifically focusing on the role of critical thinking in a student's academic life. The readings expanded upon the topic of critical thinking by describing its fundamental components which include Asking Questions, Reasoning it Out, Believing the Results as well as the obstacles arising from egocentrism and all or nothing thinking. Business analysts need to adopt structured thinking techniques like SEE-I to reach deeper problem solutions that address fundamental issues instead of superficial symptoms.

Activity:

SEE-I Analysis

S: State It

The intentional process of analysing, assessing and synthesising data in order to make well-informed decisions regarding academic challenges is known as critical thinking in problem-solving. Before making decisions regarding study techniques, assignment approaches or career choices, it entails challenging presumptions, looking at the evidence and taking into account various viewpoints.

E: Elaborate

Critical thinking makes the process of students addressing academic problems more different from the superficial mere acceptance of knowledge. Unlike mere memorization of facts or obeying orders, critical thinking asks students to pose questions of "why" and "how" behind their process of learning more aggressively.

These include three most critical areas, which we discussed in our lecture posing key questions to the problem, examining systematically potential solutions and believing and doing something about conclusions.

For students, it's about having the ability to see prejudice, question sources credibility and value the potential role of emotion in deciding what to learn or how to have assignments

completed. It's about releasing the either/or thinking of the standoff in our learning, wherein students believe there is just one right way to learn or complete assignments. Critical thinking brings with it the flexibility and resourcefulness in learning strategies.

E: Exemplify

Positive Example: When faced with a challenging group assignment, a critical thinking student would first question what makes the assignment difficult, asking the right questions. They would then gather information about team dynamics, individual strengths, project requirements and available resources, reasoning it out.

Finally, they would implement a structured plan based on this analysis, perhaps using project management tools and clear communication strategies believing and acting on results.

Contrasting Example: A non-critical thinking approach would involve immediately dividing tasks equally without considering individual skills, accepting the first solution proposed, or becoming frustrated when initial strategies don't work without analysing why they failed.

I: Illustrate

When the student uses critical thinking to solve a problem, it is similar to the detective looking for a tough case to crack. A critical thinking student considers numerous "clues" sources of data, viewpoints and solutions before making a conclusion, just like a detective will not take the first explanation offered.

Just as the students are encouraged to question sources for credibility, challenge assumptions and explore alternative solutions to problems, the same does the detective in the form of systematic techniques for gathering clues, questioning the credibility of witnesses and examining alternative explanations.

Students can utilize models like SEE-I to structure their thinking and make sure they do not omit vital information of their analysis, like the case file of a detective that organizes logically the evidence.

Peer Reviews We Got from:

Botao Ying

26 Oct 0:42| Last edited 26 Oct 0:42

Reply from Botao Ying

Did the work describe and highlight the chosen topic: Yes
Do you recommend any changes: Yes
Maybe trim the elaboration to 2 short paragraphs and pick a single, concrete student problem so the example feels more immediate.

Vignesh Karthik

26 Oct 22:24| Last edited 26 Oct 22:24
Reply from Vignesh Karthik

1. Did the work describe and highlight the chosen topic: Yes

2. Do you recommend any changes: No

3. If yes what are they? If no why?

The SEE I structure is expertly followed for the topic of Critical Thinking in Problem-Solving. Each section connects logically and builds a comprehensive understanding:

The analysis expertly follows the SEE Structure for critical thinking in problem solving also the state section provides a clear concerns definitions by the way elaborate effectively expands on this by contrasting critical Thinking.

Peer Reviews We Gave to NIBRAS SABAB:

Dhimanshi Piripun Jayasena

24 Oct 15:55

Reply from Dhimanshi Piripun Jayasena

Reply from Dhimanshi, Madhavi, Hari, Shashank, and Sankeerth,

1. **Did the work describe and highlight the chosen topic:** Yes
2. **Do you recommend any changes:** No
3. **If no, why?**
   The SEE-I analysis was clear and well-structured. It explained time management simply, gave good examples, and used a helpful comparison. The references were also included properly. Great job!

Reflections:

This week, I invested approximately 8 hours overall, dedicated to the readings, the team discussion, and drafting the SEE-I analysis.

The material on Critical Thinking provided a formal structure for what I have always considered an intuitive part of my professional life. As a full stack developer at Infosys and TCS, my work often involved debugging complex systems and optimizing performance, which requires a persistent, questioning approach a practical form of critical thinking. For instance, when a microservice failed, I had to move beyond the initial error message (the uncritical answer) and systematically ask questions about the root cause (the reasoning it out phase).

The SEE-I framework proved to be an excellent tool for formalizing this process. Applying it to the scenario of a student facing a challenging group assignment allowed me to connect the abstract concept of critical thinking to concrete actions. My team's analysis highlighted that a critical thinking student would not just divide tasks (the non critical approach) but would first question the difficulty, reason out a plan based on team strengths, and finally act

on the results. The analogy of the critical thinking student as a detective (the 'Illustrate' part of SEE-I) resonated strongly with my experience in systems analysis, where I had to gather 'clues' (data, logs, user feedback) to solve a tough technical case.

The three parts of critical thinking which include Asking the Questions and Reasoning it Out and Believing the Results have a direct impact on my professional work activities. The method establishes a formal procedure which starts with problem detection and ends with solution implementation and knowledge acquisition. The ability to overcome 'all-or-nothing thinking' after a solution fails will help me create durable cloud applications and APIs.

Conclusion:

The key outcome of Week 8 was the formal introduction and application of Critical Thinking as a structured problem-solving methodology, specifically through the SEE-I framework. This provided a valuable tool for analyzing problems systematically, moving beyond uncritical assumptions, and ensuring that solutions are based on sound reasoning a fundamental skill for any effective business analyst or software engineer.

References:

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- Lutton, E. (2025). INFO6050 Process Analysis and Problem Solving: Week 8 - Critical Thinking. University of Newcastle.
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- Paul, R., & Elder, L. (2006). The Miniature Guide to The Foundation for Critical Thinking. The Foundation for Critical Thinking.
- Snyder, S. (2016, September 22). How to Ask a Smart Question. Retrieved from https://leadingwithquestions.com/personal-growth/how-to-ask-a-smartquestion/
- Zulfiqar, S. (2018, June 30). Framework of critical thinking. Academia.edu. Retrieved from https://www.academia.edu/framework-critical-thinking

Introduction:

The week was centered on Critical Thinking 2, which is a direct extension of what was covered in week 8. The presentation focused on reasoning, incorporating the elements of reasoning including purpose, question at issue, information, concepts, assumptions, inferences, implications, and point of view and the standards of thinking like clearness, accuracy, relevance, importance, sufficiency, depth, breadth, and precision. We spent time thinking about thinking, reflecting on our own reasoning to produce valid reasoning. The lab session engaged us in readings on critical thinking strategies, activities in class to apply the elements of reasoning and distinguish facts from opinions, and both the completion of peer reviews for the Week 8's forum posts, and optionally case studies. All these ideas fall in line with the course's strong focus on analytical tools for researching business processes and making informed decisions.

Reflections:

The material this week deepened my understanding of critical thinking as a structured, intentional practice rather than a casual skill. Exploring the reasoning wheel and standards clarified how small gaps like unclear purpose or unchecked assumptions can weaken decisions. In my personal full stack development practice, I often diagnose issues in code or system performance, but now I see the value in pausing to check breadth and depth before jumping to fixes. This prevents quick patches that later fail under load.
The fact or opinion activity was simple but powerful. Statements like “The census is a waste of resources” sound factual but are opinions without evidence. This reminded me to challenge vague claims in project planning or requirements, asking “What data supports this?” leads to clearer outcomes. Using M365 Copilot with the prompt on decision making gave a structured response that echoed my own thoughts, breaking problems into elements reduces bias and improves accuracy. It felt like a digital mirror for self review.

Peer reviews from Week 8 were helpful too. Giving feedback sharpened my eye for precision in others’ work, and receiving it encouraged me to quantify impacts in my examples like showing how a delay affects users, not just stating it. This back and forth reflects the 8 Ways, community learning, non-linear rethinking, and reconstructing ideas. On a personal level, I applied this to a small app I’m building for tracking study progress. I assumed one approach was best, but after applying the elements, I tested alternatives and found a simpler, faster solution. The week connected technical problem solving with business level clarity, which feels essential for moving into more strategic IT roles in Australia. No real difficulties just steady growth in confidence and awareness.

Conclusion:

Key outcomes included a solid grasp of reasoning elements and standards, practical reinforcement through lab tasks and AI interaction, and stronger ability to apply critical thinking in both personal projects and study. This prepares me well for collaborative analysis and innovation in the coming weeks.

References:

- Foundation for Critical Thinking. (n.d.). The elements of reasoning and the intellectual standards. Retrieved November 15, 2025, from https://www.criticalthinking.org/pages/the-elements-of-reasoning-and-the-intellectual-standards/480
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Week 10:

Introduction:

This week shifted to Group Critical Thinking, building on individual reasoning skills. The lecture covered creative group techniques like the 9 dot puzzle to challenge assumptions, Edward de Bono's Six Thinking Hats for balanced perspectives including white: facts, red: emotions, black: risks, yellow: positives, green: ideas, blue: overview, the Delphi method for anonymous expert input, and strategies to counter groupthink such as brainstorming and nominal group technique. It also introduced decision tools like risk analysis and impact assessment. The lab focused on readings about rich pictures for visual process mapping and stakeholder management, a class discussion on how goals shape thinking and the "thinking outside the box" concept from the article, and time for the case study report with checklist reminders on structure, APA referencing, image citations, stakeholder ranking, process identification, and linking next steps. These supported collaborative problem solving in business contexts.

Reflections:

This week's emphasis on group dynamics showed how structured methods can turn potential pitfalls like conformity into strengths for innovation. The Six Thinking Hats stood out as a practical way to ensure all angles are covered something I can simulate even in solo work by cycling through each hat. In my personal full stack development practice, when planning a feature, I might start with white hat facts like requirements, then green for creative options, avoiding early black hat negativity that kills ideas. The Delphi technique's iterative, blind feedback appeals for reducing ego in teams, much like code reviews in open source projects where anonymity could encourage honesty.

Discussing goals and decision making revealed how fixed objectives create invisible boundaries, as the article explained with the cliché of "thinking outside the box." We know the idea but stick to safe paths due to comfort or pressure, the 9 dot solution requires ignoring self made rules. This encouraged me to loosen goal constraints in my practice sessions, instead of aiming for a specific outcome, I explored unrelated tools, leading to a more efficient database query I hadn't considered. Rich pictures seem useful for sketching messy problems visually, helping deconstruct complexity non verbally per the 8 Ways, while stakeholder grids build on earlier analysis by prioritizing engagement. Personally, applying this to a study tracker app I'm building, I used a mental power/interest map for "users" like myself and future sharers, adjusting features based on impact. It all ties technical creativity to business strategy, vital for IT roles in Australia, with smooth integration of concepts and no challenges.

Conclusion:

Key outcomes were mastering group tools like Six Hats and Delphi to foster diverse thinking, identifying ways goals limit creativity and how to break free, and practical steps for case study refinement, enhancing readiness for team driven process solutions ahead.

References:

- Anderko, M. (n.d.). *Thinking outside the box cliché*. Retrieved November 15, 2025, from https://mutualink.net/thinking-outside-the-box-cliche-we-understand-it-but-do-not-practice-it/
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Introduction:

This week centered on innovation and sustainability in information systems, tying into process analysis through Green IT practices. The lecture discussed harnessing IT for environmental impact, including principles like energy efficient design, e waste reduction, and emerging tech greening. It highlighted global challenges from the E waste Monitor, such as rising electronic waste volumes and innovation needs for circular economies. The lab involved reviewing readings on e-waste innovation, Green IT techniques, and a class discussion in small teams to identify two examples each of the four innovation types from Dieffenbacher's model: incremental , architectural , disruptive, and radical. This built toward the forum activity on course learnings, with time for exam preparation, aligning with the course's push for reflective, innovative problem solving in business contexts.

Activity:

What was the main skill or knowledge you gained while completing this course?

The primary skill I developed in INFO6050 is that I am now capable of applying critical thinking in business process analysis, and systematically examine problems, model solutions using UML, and innovate information systems. I learned definitions including critical thinking as reasonable, reflective thinking focusing on what to do, metacognition as a key feature of critical thinking, and standards of evaluation such as accuracy and relevance. The part called asking questions that probe, reasoning logically, and acting purposefully on results not only changed the way that I cope with biases but also evaluate multiple options, such was displayed in the SEE-I framework.

What processes would you suggest to a student that was enrolling in this course next year?

For students enrolling next year, I suggest adopting the Business Analysis Process Model, start by investigating situations with qualitative techniques like interviews and workshops, then perform stakeholder analysis using tools like the stakeholder wheel and RACI matrices to identify influences. Model processes with UML activity and use case diagrams to visualize flows. Integrate critical thinking to analyze needs and evaluate feasibility, and foster innovation by creating a culture through intentional intent, unstructured time, and meaningful metrics. Participate in labs and discussions for practical application, the 8 Ways pedagogy enhances holistic learning.

Reflections:

This week wrapped up the course with a forward looking lens on sustainability and innovation, prompting me to connect earlier tools like UML modeling with broader impacts like reducing e waste through smarter IT designs. The readings on Green IT principles, such as virtualization for energy savings and lifecycle assessments, made me rethink how processes can be eco friendly for instance, in my personal full stack practice, I started auditing code for efficiency to mimic modular hardware that eases recycling, as suggested in the E waste Monitor's chapter on improvements. The class discussion on innovation types was engaging, our team brainstormed examples like incremental updates in app patches low risk, steady gains versus radical shifts like AI driven predictive maintenance creating new markets. It

highlighted how low-tech/high-market innovations, like sustainable packaging for devices, solve real problems without overhauling tech stacks.

The forum activity felt like a natural culmination, forcing non linear reflection per the 8 Ways I deconstructed my journey from basic fact finding to innovative modeling, then reconstructed advice with storytelling elements, like sharing how SEE-I helped unpack biases in a practice project. Personally, it inspired me to apply radical innovation to my study tracker app by integrating green metrics, such as tracking carbon footprints of cloud usage via simple APIs. This links technical skills to ethical business analysis, crucial for strategic IT paths in Australia where sustainability regulations are tightening. No sticking points; it just solidified the course as a bridge from reactive fixing to proactive, thoughtful design.

Conclusion:

Key outcomes were insights into Green IT for sustainable processes, practical examples of innovation types for problem-solving, and a synthesized view of course skills like critical thinking and modeling, leaving me equipped to advise peers and innovate responsibly in future work.

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Week 12:

Introduction:

This final week served as a course capstone, emphasizing career transition and consolidation. The lecture reviewed key themes like process modeling, critical thinking, and innovation, with a forward look at applying them in professional settings. The lab directed focus to exploring Business Analyst job vacancies on Seek and Indeed to connect academic skills to real-world opportunities, finalizing Assessment 2 by ensuring the Week 11 forum post adhered to the required structure, and preparing for the exam through analyzing sample questions, rewriting them for clarity, mapping coverage to weekly tasks like UML diagrams, stakeholder analysis , and noting the exclusive use of UoN M365 Copilot for any AI assistance. This reinforced the circular learning loop, where assessments inform exam responses, aligning with the course's holistic approach to problem solving.

Reflections:

Wrapping up the trimester felt reflective and motivating, especially browsing the job sites which brought the course's practical value into sharp focus. On Seek, listings like the SaaS tech Business Analyst role in a fast-growing firm highlighted needs for software implementations and customer collaboration, while the hybrid Brisbane position stressed enterprise transformation skills directly from our UML modeling and Agile discussions. Indeed showed similar demands, such as process documentation in global finance projects, echoing stakeholder engagement and BPR techniques. Common threads across roles included data analysis, efficiency improvements, and cross-team work, often requiring 3-5 years' experience but open to IT Master's grads with strong analytical foundations. Salaries ranged from $120k-$130k annually to $800-$900 daily contracts, realistic for mid level entry in Sydney or Brisbane, which excites me given my full stack practice.

This tied beautifully to my initial expectations of bridging technical expertise with business strategy; I've evolved from viewing processes as code flows to seeing them as stakeholder driven ecosystems, thanks to tools like RACI matrices and Six Thinking Hats. In my personal development sessions, I now apply critical thinking standards—like checking assumptions in a recent app refactor to simulate BA tasks, such as mapping user journeys with activity diagrams. The exam review prompted non linear rethinking per the 8 Ways, deconstructing sample questions on innovation types and reconstructing answers with storytelling from rich pictures, creating a learning map of the entire journey. Finalizing the journal uncovered how peer reviews fostered community, much like job collaboration, and inspired me to tailor my resume with course-specific examples, like using SEE-I for requirements elicitation. Personally, it eased transition anxieties for Australian IT roles, where sustainability from Green IT is increasingly key. Smooth sailing overall, with a sense of closure and momentum.

Conclusion:

Key outcomes encompassed targeted job market insights validating course relevance, polished Assessment 2 submission, and strategic exam readiness through content mapping, positioning me confidently for professional and academic next steps.

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Final Conclusion:

Process Analysis and Problem Solving delivered a comprehensive, expertly structured curriculum guided by Professors with exceptional clarity and enriched by Tyson Yunkaporta's 8 Ways Aboriginal pedagogies that transformed foundational business analysis into advanced, strategic problem solving. The course progressed seamlessly from defining business analysis, systems thinking, and design thinking in Week 1, to hands on UML modeling of airport check in processes in Week 2, and PESTLE analysis of Mahindra's EV and tractor strategy in Australia in Week 3. Week 4's VMOST alignment using Myer and Kmart, Week 5's stakeholder power/interest grid for an offshore wind farm, and Week 6's precise application of functional and non functional requirements with verification and validation in a drone disaster system built rigorous, real-world process documentation and quality assurance skills.

From Week 7 onward, the course deepened critical thinking through UML modeling of ore crushing workflows, SEE-I framework analysis of academic challenges, reasoning elements and standards to separate facts from opinions, and group innovation tools like Six Thinking Hats and Delphi to counter groupthink. Innovation types, Green IT principles for e-waste reduction and energy efficient design, and rich pictures for visual mapping culminated in Week 11's synthesis of critical thinking with sustainable process innovation and Week 12's alignment with Australian job market demands via Seek and Indeed. Through consistent peer reviews, reflective journaling, and collaborative forums, the course fostered non linear, community driven learning. This robust toolkit now equips me to model resilient workflows, lead stakeholder centered change, and design sustainable, innovative IT solutions directly enhancing my future effectiveness in cloud application development, API design, and strategic business analysis roles in Australia's evolving digital landscape.